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X_L : cells that are exposed to light
X_S : cells present in shaded region

$$N_L = \left[c_1 + (c_2 - c_1) \times r_1 \right] \times N \quad - (1)$$

$$N_S = N - N_L \quad - (2)$$

$$X_L = \{ x_1, x_2 \ldots\ldots x_{N_L} \} \quad - (3)$$

$$X_S = \{ x_{N_L+1}, x_{N_L+2} \ldots\ldots x_N \} \quad - (4)$$

$$\alpha = e^{-1/T} \quad - (5)$$

$$X_{L_{new1}} = X_{rand} + \alpha \beta r_2 \odot \left| X_{rand} - X_L^i(t) \right| + \alpha \beta r_3 \odot \left| X_{L_{best}}^i(t) - X_L^i(t) \right| \quad - (6)$$

$$X_{L_{new2}}(t) = X_L^i(t) + \alpha r_4 \odot \left| X_{L_{best}}^i(t) - X_L^i(t) \right| \quad - (7)$$

$$r_4 = 2 \times rand() - 1$$

$$X_{S_{new1}} = X_S^i(t) + \alpha \beta r_5 \odot \left| X_S^i(t) - X_{rand} \right| \quad - (8)$$

$$X_{S_{new2}} = X_{L_{rand}} + \alpha \times \beta \times r_6 \odot \left| X_{L_{best}} - X_S^i(t) \right| \quad - (9)$$

$$\text{Curvature} = \beta \times \left(\alpha - \frac{\text{Mean_fitness}(X_L)}{\text{Best_fitness}(X)} \right) \quad - (10)$$

$$FOC = r_7 \times \text{Curvature} \odot \left(x^i(t) - X_{Best}(t) \right) \quad - (11)$$

$$\text{cell vicinity} = \alpha \times \beta \times r_8 \odot \left(\frac{x^i(t) + x^{i+1}(t)}{2} \right) \quad - (12)$$

$$\cancel{X_{new}^i(t) = x^i(t) + \varepsilon_7 \times \text{curvature} \odot}$$

$$X_{new}^i(t) = x^i(t) + FOC + \text{cell vicinity}$$

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Step1: Initialize population

$$X = \{x_1, x_2, x_3 \ldots x_N\}$$

N: no. of cells

Step2: Evaluate fitness function for each cell

$$F(x) = [f_1(x), f_2(x), f_3(x) \ldots f_M(x)]$$

$$F = \begin{bmatrix} f_1(x_1) & f_2(x_1) & f_3(x_1) & \ldots & f_M(x_1) \\ f_1(x_2) & f_2(x_2) & \ldots & & f_M(x_2) \\ \vdots & & & & \end{bmatrix}$$

: $N \times M$

M: no. of objectives

Step3: we will rank all the vectors using non-dominated sorting

Pareto fronts: $[F_1, F_2 \ldots F_K]$

Ranks: $[r_1, r_2, \ldots, r_N]$

where r_i is rank of F_i

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Step 4: Calculate crowding distance

$$CD_i = \sum_{m=1}^{M} \frac{f_m(i+1) - f_m(i-1)}{f_m^{max} - f_m^{min}}$$

$x_i < x_j$

if $rank_i < rank_j$

or

$rank_i = rank_j \quad CD_i > CD_j$

Now we get the initial pareto front

Step 5: Divide the population X into X_L & X_S using Eq. (1), (2), (3) & (4)

$$X_L = \{x_1, x_2 \dots x_{N_L}\}$$
$$X_S = \{x_{N_L+1}, \dots x_N\}$$

$$X = \begin{bmatrix} X_L \\ X_S \end{bmatrix}$$

Step 6: Evaluate fitness function for X_L & X_S

$$F_L = \begin{bmatrix} f_1(x_1) & f_2(x_1) & \dots & f_M(x_1) \\ f_1(x_2) & \dots & & f_M(x_2) \\ \vdots & & & \vdots \\ f_1(x_{N_L}) & \dots & & f_M(x_{N_L}) \end{bmatrix}$$

$$F_S = \begin{bmatrix} f_1(x_{N_L}) & f_2(x_{N_L}) & \dots & f_M(x_{N_L}) \\ f_1(x_{N_L+1}) & f_2(x_{N_L+1}) & \dots & f_M(x_{N_L+1}) \\ \vdots & & & \vdots \end{bmatrix}$$

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Step 7: find X_{Lbest} + X_{Sbest}

Repeat these steps until condⁿ met:

1) Compute α : $\alpha = e^{-t/T}$

2) for(int $i = 1$ to N_L):

Each parent cell $\xrightarrow{\text{Mitosis}}$ Daughter cell 1 + Daughter cell 2

Daughter cell 1: mutation operator

$$X_{Lnew1}(t) = X_{rand} + \alpha\beta r_2 \odot |X_{rand} - X_{iL}(t)| + \alpha\beta r_3 \odot |X_{iL_{best}}(t) - X_{iL}(t)|$$

Daughter cell 2 : Auxin redistⁿ

$$X_{Lnew2}(t) = X_{iL}(t) + \alpha r_4 \odot |X^{i}_{Lbest}(t) - X_{iL}(t)|$$

3) Merge it into population:

$$X_L = \begin{bmatrix} X_L \\ X_{Lnew1} \\ X_{Lnew2} \end{bmatrix} : 3N_L$$

4) for($i = 1$ to N_S): Each parent $\longrightarrow$ Daughter cell 1 + Daughter cell 2

$$X_{Snew1}(t) = X_{iS}(t) + \alpha\beta r_5 \odot |X_{iS}(t) - X_{rand}|$$

$$X_{Snew2}(t) = X_{Lrand} + \alpha\beta r_6 \odot |X_{Lbest} - X_{iS}(t)|$$

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5) Merge into population :

$$X = \begin{bmatrix} X_L \\ X_S \\ X_{LNew1} \\ X_{LNew2} \\ X_{SNew1} \\ X_{SNew2} \end{bmatrix} \quad ; \quad 5N$$

6) for (i = 1 to N) :

$$X_{inew}(t) = X^i(t) + FOL + Cell_Vicinity$$

7) Merge into population :

$$X = \begin{bmatrix} X_L \\ X_S \\ X_{LNew1} \\ X_{LNew2} \\ X_{SNew1} \\ X_{SNew2} \\ X_{New} \end{bmatrix} \quad ; \quad 6N$$

8) Evaluate fitness of all the cells.
9) Apply Non dominated sorting
10) Compute crowding distance
11) Select best N cells

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